

Assessment Specification – Portfolio Task 4

Module Details

Module Code	UFCE8B-30-3
Module Title	Data Science for Cyber Security
Module Leader	Ahsan Ikram
Module Tutors	Ahsan Ikram, Andrew McCarthy
Year	2025-26
Task	04
Total number of assessments for this module	01
Weighting	20%
Date issued to students	16/02/2026

Task Specification

You are a cyber security analyst consulting for a financial organisation.

Their recent risk assessment suggests a higher than usual threat of a malware attack. They have decided to be proactive and improve their defences.

One of the areas they are focussing on is the development of a real-time visualisation dashboard. In the first prototype of the data visualisation dashboard, they aim to monitor,

- 1) Network Traffic (throughput, protocol, source etc.)
- 2) Endpoints (current issues, Operating System (OS) version, firewall status etc.)
- 3) Data backups (backup coverage, backup encryption, backup validation etc.)
- 4) Staff readiness and awareness of cyber threats (pending training, job role risk profile, current risk profile etc.)

Your task is to advise the local government on what data should be collected and based on that, next, preparation of the dashboard. A breakdown of tasks and marks is shown below.

Sample Files: No files were given for this task.

Deliverable: You are required to submit,

1. A document that contains your suggestions on the details of the data that should be collected.
The table below provides some examples.

Category	Sub-category	Metric / Format	Frequency	Collection Mode	Collection Mechanism	Recorded at	Storage
Network	Protocol	Text	Real-time	Automated	Network tap	Gateway router	CSV file
Endpoint	Staff PC	Text	10 min	Automated	SNMP trap	Host machines	CSV file
Staff readiness	Current Profile	3-scale (low-medium-high)	3 months	Manual	HR records	HR system	Excel file

2. A cyber security dashboard prototype (created in Python).
 - a. You don't have to collect or use actual data for this task. You can generate a small set (5-10 rows) of dummy data.
 - b. The dashboard should use at least two inputs. For example, one CSV (Comma Separated-Values) or text or excel file or a database table for network traffic, one CSV/text/Excel file for endpoint data.
 - c. Focus on the usability and user experience aspects of the dashboard; in terms of the selection of the layout, selection of graphs, placement of important information, selection of fonts/colors/separators etc.

Marking criteria:

	Task	Marks
1	Three identified data metrics each, for network traffic monitoring, endpoints monitoring, data backups monitoring and staff readiness monitoring.	4
2	Prototype dashboard using at least two input files and six charts.	8
3	At least one geo-spatial visualisation (included in the six charts in part-2).	2
4	At least one interactive visualisation (included in the six charts in part-2). Choose an appropriate interaction, for example, data filter, layout change, auto-refresh etc.	2
5	Effective layout and design of the dashboard.	4
	Total	20